Phonon Assisted Absorption

A Non-perturbative Approach

Yifan Liu, Vivek Aji

University of California, Riverside

QuVET

Overview

Motivation

Model
- Model Hamiltonian
- Shrieffer-Wolff/Polaron Transformation as One-step RG

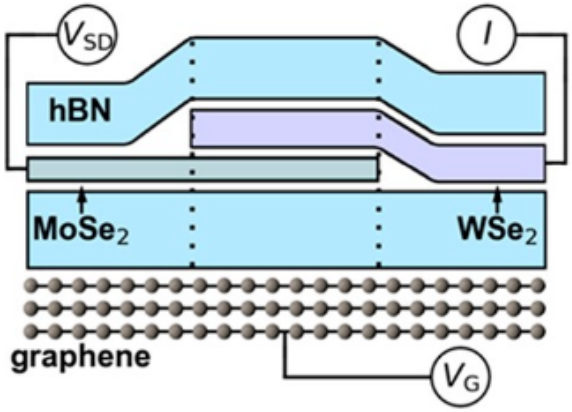
Absorption Spectrum
- Absorption and Retarded Green's Function

Results

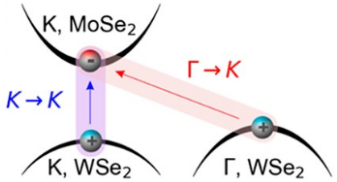
Discussion

Motivation

Recent experiments [Barati et al., Nano Lett. 2022] demonstrate that vibronic structures can be probed and electrically controlled in vdW heterostructures.

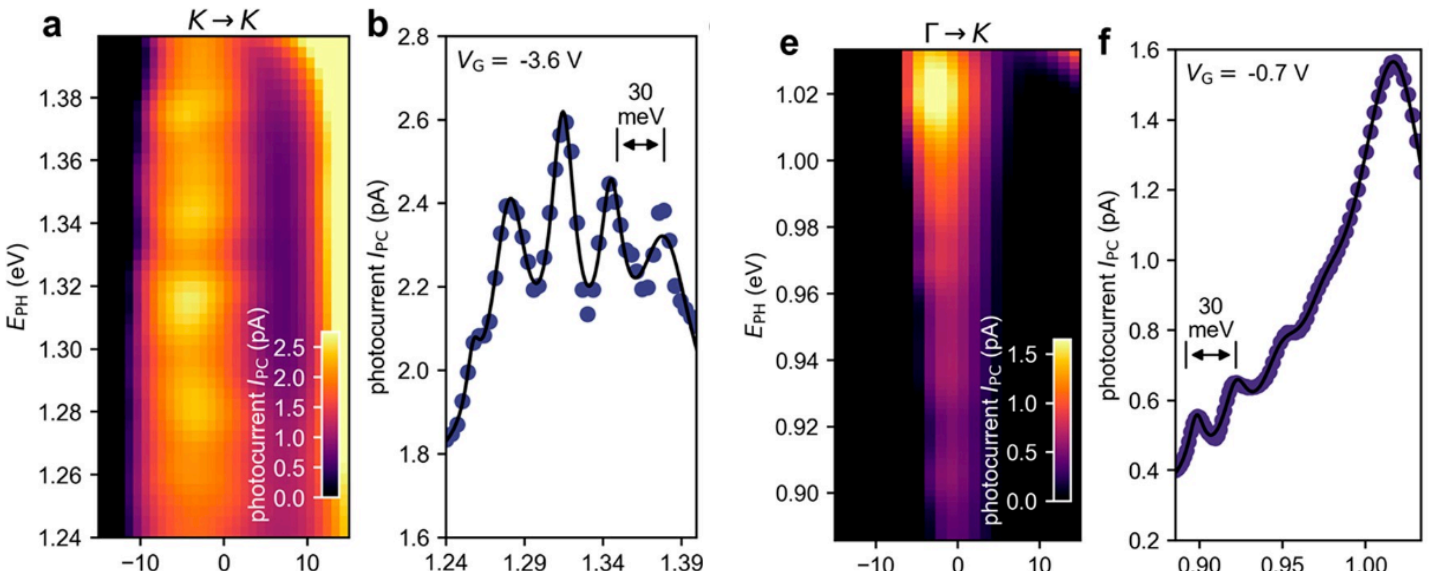


- 2L-WSe_2/$MoSe_2$: K→K (direct) and Γ →K (indirect) interlayer excitons (IXs).



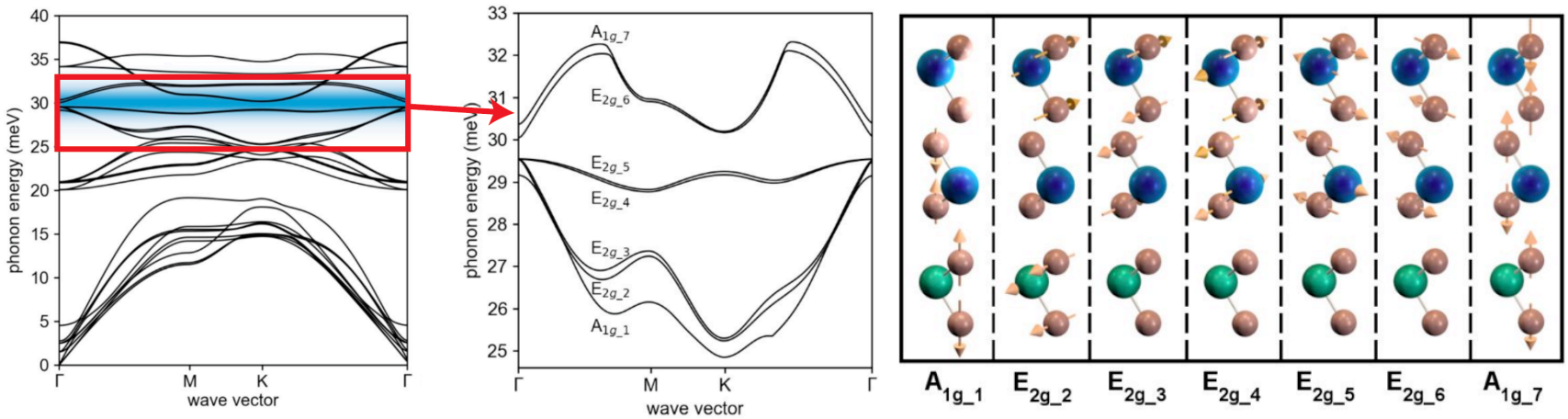
- Photocurrent spectroscopy probes pronounced periodic sidebands, interpreted as **vibronic coupled IX-phonon states**.

Q1: How does electron-phonon coupling produce such periodic sharp features?



Motivation

Q2: What is the phonon modes responsible for these peaks?



[Barati et al., 2022]

- Phonon replicas show nearly periodic spacing ~ 30 meV. $\rightarrow$ DFT calculations identify 7 nearly flat **optical phonon** branches near this energy.

Model Hamiltonian

Consider a minimal two-band model. The full Hamiltonian

$$H = \underbrace{H_e^0 + H_{ph}^0 + H_{em}^0}_{H_0} + \underset{Coulomb}{H_{e-e}} + \underset{Electron-phonon}{H_{e-ph}} + \underset{Light-matter}{H_{e-em}} :$$

$$H^0 = \sum_{\mathbf{k}\lambda} \epsilon_{\mathbf{k}\lambda} a^\dagger_{\mathbf{k}\lambda} a_{\mathbf{k}\lambda} + \sum_{\mathbf{q}\alpha} \Omega_{\mathbf{q}\alpha} D^\dagger_{\mathbf{q}\alpha} D_{\mathbf{q}\alpha} + \sum_{\mathbf{q}} \omega_{\mathbf{q}} B^\dagger_{\mathbf{q}} B_{\mathbf{q}}$$

$$H_{e-e} = \frac{1}{2} \sum_{\mathbf{k}\mathbf{k}'\mathbf{q}\lambda\lambda'} V_{\mathbf{q}} a^\dagger_{\mathbf{k}-\mathbf{q}\lambda} a^\dagger_{\mathbf{k}'+\mathbf{q}\lambda'} a_{\mathbf{k}'\lambda'} a_{\mathbf{k}\lambda}$$

$$H_{e-ph} = \sum_{\mathbf{k}\mathbf{q}\lambda\lambda'\alpha} \Omega_{\mathbf{q}\alpha} g^{\lambda\lambda'}_{\mathbf{q}\alpha} a^\dagger_{\mathbf{k}-\mathbf{q},\lambda'} a_{\mathbf{k}\lambda} (D_{-\mathbf{q},\alpha} + D^\dagger_{\mathbf{q}\alpha})$$

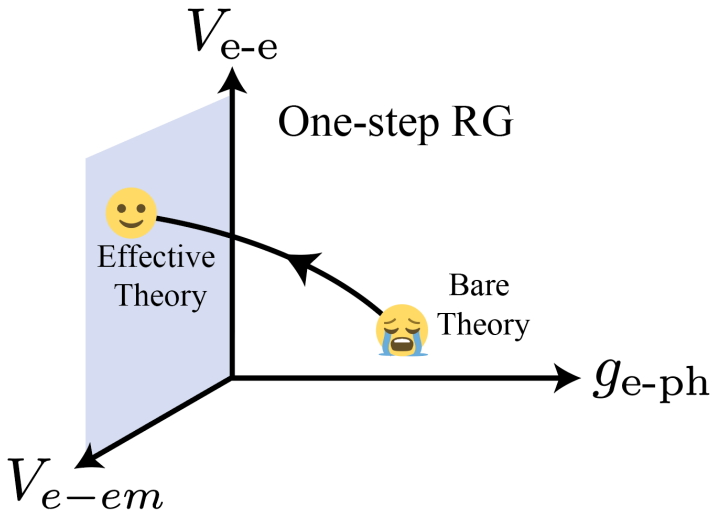
$$H_{e-em} = -\sum_{\mathbf{k}\mathbf{q}\lambda} i\mathcal{F}^{\lambda\bar{\lambda}}_{\mathbf{q}} a^\dagger_{\mathbf{k}+\mathbf{q}\lambda} a_{\mathbf{k}\bar{\lambda}} B_{\mathbf{q}} + h.c..$$

Key features:

- Two-band model $\lambda \in \{c, v\}$ (conduction/valence)
- Optical phonon approximation $\Omega_{\mathbf{q}\alpha} \approx \Omega_\alpha$.
- Electron-phonon coupling matrix $g^{\lambda\lambda'}_{\mathbf{q}\alpha} \sim g_1 \sigma_1^{\lambda\lambda'} + g_3 \sigma_3^{\lambda\lambda'}$.

Shrieffer-Wolff/Polaron Transformation as One-step RG

The idea of polaron transformation is to perform a canonical transformation $U = e^S$ so that the H_{e-eh} in the transformed Hamiltonian $e^{\mathrm{ad}_S} H = e^S H e^{-S}$ vanishes, while remaining terms are renormalized.
$S = \sum \mathcal{C}_{k\lambda,k'\lambda'} a^\dagger_{k'\lambda} a_{k\lambda}$ and $\mathcal{C} \sim g * (D - D^\dagger)$.



Schematic RG flow

- The polaron transformation allows "integrating out" phonon degrees of freedom.
- No higher order terms are generated as $\mathrm{ad}^n_S H_{e-ph} = 0$ for $n > 1$.
- Captures effective low-energy theory leading to phonon-assisted transitions.

After the transformation: electron $\rightarrow$ polaron (electron dressed by phonons) and

$$\bar{H} = \underbrace{H^0_{pol} + H^0_{ph} + H^0_{em}}_{\bar{H}_0} + \underset{Coulomb}{H_{pol-pol}} + \underset{Polaron-phonon}{\cancel{H_{pol-ph}}} + \underset{Light-matter}{H_{pol-em}}$$

Absorption and Retarded Green's Function

- Optical absorption is calculated via the Green's function formalism:

$$\alpha(\omega) \propto -\mathrm{Im} P^R_>(\omega)$$
$$P^R_>(t) = -i\frac{\Theta(t)}{N_{\mathrm{cell}}}\sum_{\mathbf{k}\mathbf{k}'}\langle a^\dagger_{\mathbf{k}v}(t)a_{\mathbf{k}c}(t)a^\dagger_{\mathbf{k}'c}(0)a_{\mathbf{k}'v}(0)\rangle$$

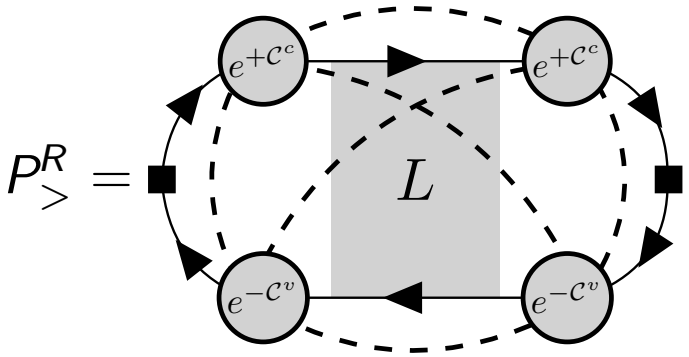
- Given that $a^\dagger \to e^{\mathcal{C}}a^\dagger$, the transformed greater four-point correlation function

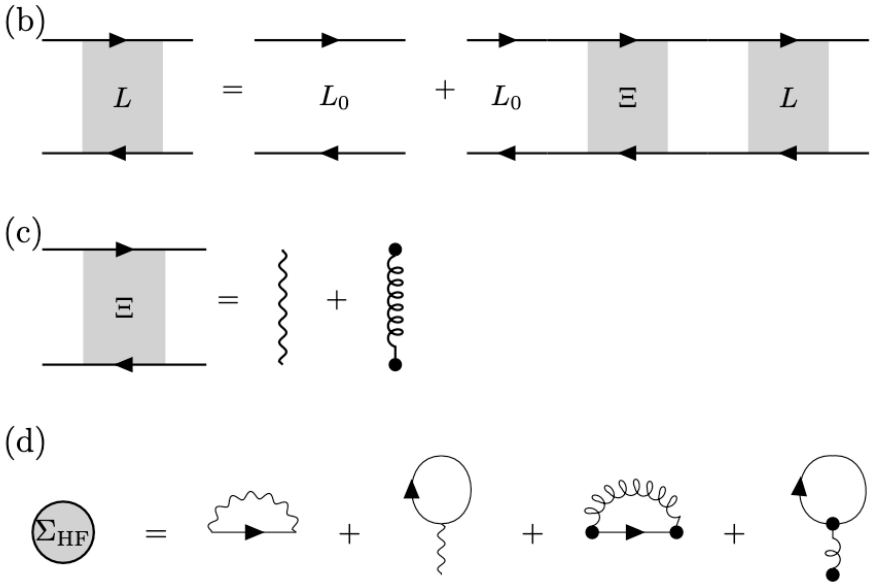
$$P^R_>(t) = -i\frac{\Theta(t)}{N_{\mathrm{cell}}}\sum_{\mathbf{k},\mathbf{k}',\{\mathbf{k}_i\lambda_i\}}\left\langle a^\dagger_{\mathbf{k}_1\lambda_1}(t)a_{\mathbf{k}_2\lambda_2}(t)a^\dagger_{\mathbf{k}_3\lambda_3}(0)a_{\mathbf{k}_4\lambda_4}(0)\right\rangle$$
$$\times\left\langle (e^{\mathcal{C}(t)})_{\mathbf{k}v,\mathbf{k}_1\lambda_1}(e^{-\mathcal{C}(t)})_{\mathbf{k}_2\lambda_2,\mathbf{k}c}(e^{\mathcal{C}(0)})_{\mathbf{k}'c,\mathbf{k}_3\lambda_3}(e^{-\mathcal{C}(0)})_{\mathbf{k}_4\lambda_4,\mathbf{k}'v}\right\rangle$$

- In the polaron picture, the polaron and phonon operators separate, thus the thermal averages can be performed independently!

Relation to perturbative method—Feynman Diagrams

In a special case where g_1 is turned off, $e^{\mathcal{C}(t)}_{\mathbf{k}\lambda,\mathbf{k}'\lambda'}$ is diagonal in band indices. The transformed greater four-point correlation function can be represented as the following diagram.





Results

In the absence of the coupling of phonons to interband transitions g_1, the absorption spectral function is:

$$\alpha(\omega, T) = e^{-4\widetilde{G}_g(T)} \sum_{\nu} \sum_{m=-\infty}^{\infty} \frac{e^{\frac{1}{2}m\beta\Omega}|\psi_\nu(\mathbf{0})|^2\gamma}{(\omega - m\Omega - E_{\mathbf{0}\nu})^2 + \gamma^2} I_m\left(\frac{4|g|^2}{\sinh(\beta\Omega/2)}\right)$$

where $|g| = g_3$, $\widetilde{G}_g(T) = [2n_B(\Omega, T) + 1]g_3^2$.

- At zero temperature, $I_m\left(\frac{4|g|^2}{\sin(\beta\Omega/2)}\right) \sim \frac{1}{\Gamma(m+1)}\left(\frac{g_{cv}^2}{2\sinh(\beta\Omega/2)}\right)^m \rightarrow$ the weight of the peaks given by a Poisson distributions $\sim e^{-4|g|^2}\frac{(4|g|^2)^m}{m!}$.

Results

By turning on the coupling of phonons to interband transitions g_1, the resulting spectral function consists of three contributions $\alpha(\omega, T) = \sum_{i=0}^{2} \alpha^{(i)}(\omega, T)$:

$$\alpha^{(0)}(\omega, T) = \mathcal{W}_0(g) \sum_{\nu} \frac{|\psi_\nu(\mathbf{0})|^2 \gamma}{(\omega - E_{\mathbf{0}\nu})^2 + \gamma^2}$$

$$\alpha^{(1)}(\omega, T) = \mathcal{W}_1(g) e^{-4\widetilde{G}_g(T)} \sum_{\nu} \sum_{m=-\infty}^{\infty} \frac{(-1)^m e^{\frac{1}{2} m\beta\Omega} |\psi_\nu(\mathbf{0})|^2 \gamma}{(\omega - m\Omega - E_{\mathbf{0}\nu})^2 + \gamma^2} I_m \left(\frac{4|g|^2}{\sinh(\beta\Omega/2)} \right)$$

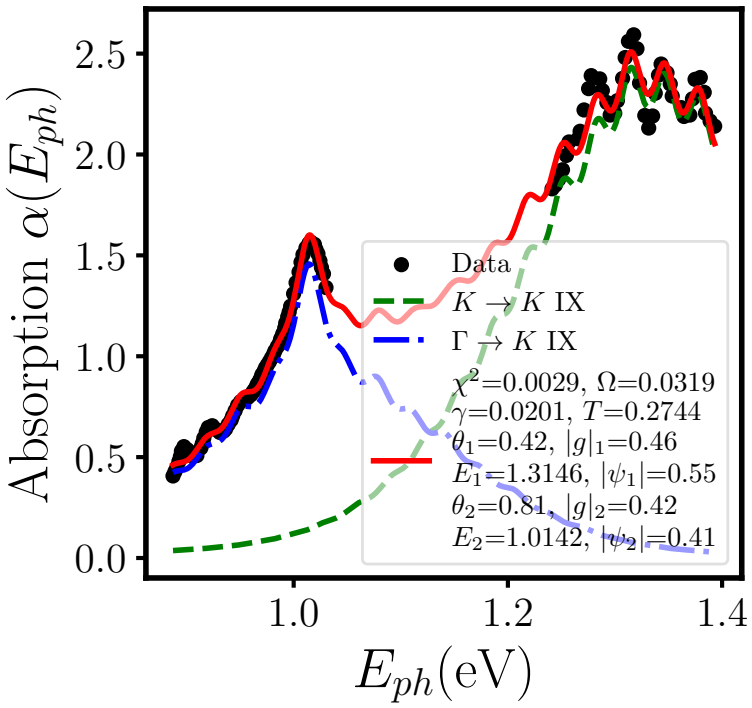
$$\alpha^{(2)}(\omega, T) = \mathcal{W}_2(g) e^{-4\widetilde{G}_g(T)} \sum_{\nu} \sum_{m=-\infty}^{\infty} \frac{e^{\frac{1}{2} m\beta\Omega} |\psi_\nu(\mathbf{0})|^2 \gamma}{(\omega - m\Omega - E_{\mathbf{0}\nu})^2 + \gamma^2} I_m \left(\frac{4|g|^2}{\sinh(\beta\Omega/2)} \right)$$

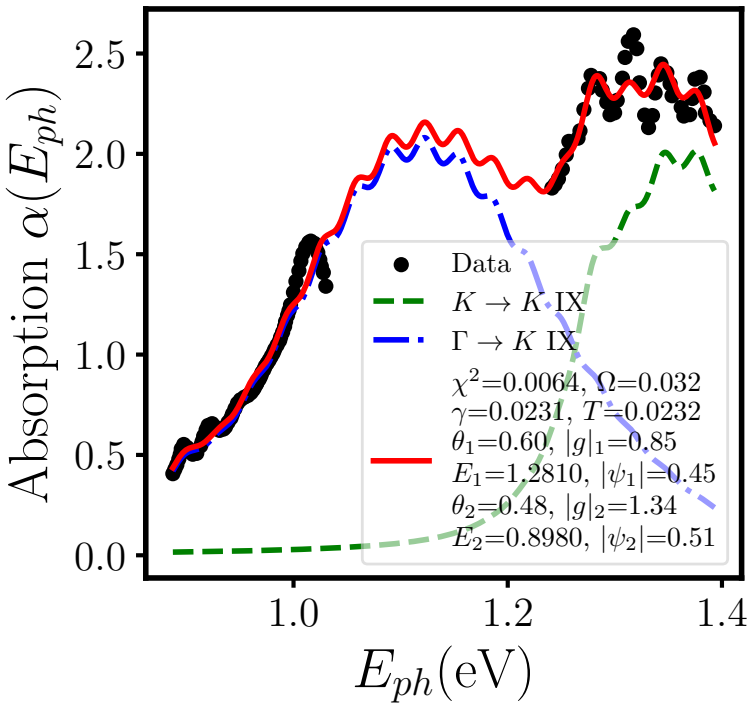
where the weight functions $\mathcal{W}_{0,1,2}(g)$ depend on coupling matrix g, and $|g| = \sqrt{g_1^2 + g_3^2}$, $\theta = \arcsin(g_1/|g|)$, $\widetilde{G}_g(T) = [2n_B(\Omega, T) + 1]|g|^2$.

- Zero-phonon peak enhancement due to $\alpha^{(0)}$.
- Partial cancellation for odd peaks and enhancement for even peaks between $\alpha^{(1)}$ and $\alpha^{(2)}$.

Results

Position of Zero-phonon peak?

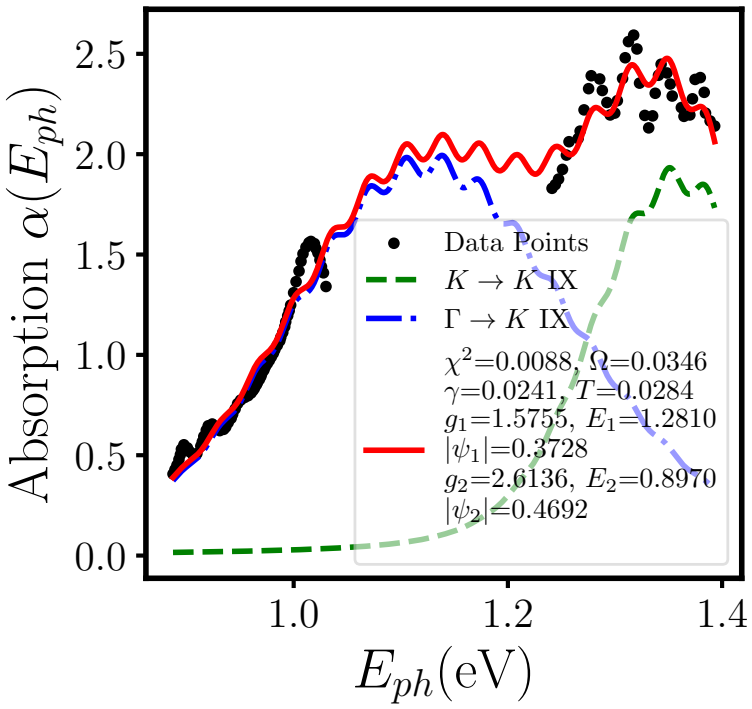


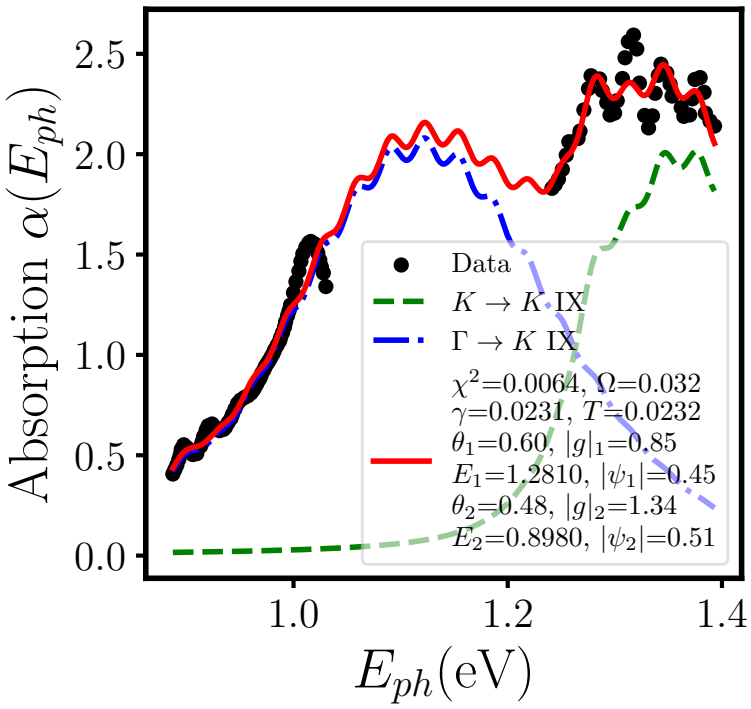


- The unconstrained fit (left), where the zero-phonon peak position is allowed to vary freely, provides a better agreement with the data but results in an unphysical temperature of $T = 0.2744$ eV.
- By constraining the zero-phonon peak position to reasonable physical range ~ 0.9eV , the fit yields a more physically reasonable temperature of $T = 0.0232$ eV(right).

Results

Effect of turning on g_1 coupling:





- Turning on the coupling of phonons to interband transitions improves the fit by $\sim$ 30%.

Discussion

- Phonon interactions play a crucial role in determining optical properties.
- The study demonstrates the significance of vibronic states in absorption spectra.
- By turning on interlayer e-ph coupling, the resulting spectral function better captures the experimental data.